Microscale Phase Change Produces Inverse Cascade in Turbulent Clouds: Evidence from DNS

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Key Points:

- Turbulence-driven direct cascade and microphysics-driven inverse cascade intersect to create a spectral break in the cloud water field.
- Across different turbulence intensities, the break occurs at a spatial wavelength of approximately eight times the Kolmogorov length.
- Continuous injection of microscale cloud water variance sustains spatial inhomogeneity and upscale variance transfer in turbulent clouds.

Abstract

The energy spectrum of cloud water content reveals a distinct break, occurring when turbulence-driven direct cascade is disrupted by another cascade. To understand the mechanism behind the second cascade, this study implements a particle-based direct numerical simulation approach to model entrainment-mixing in a turbulent cloud. All turbulence scales are explicitly resolved and cloud droplets are tracked in a Lagrangian fashion. It is shown that microscale evaporation of cloud droplets injects variance at the small scales, creating an inverse cascade in the cloud water field. The intersection of direct and inverse cascades is marked by a spectral break. The physical-space length corresponding to the break scales with the Kolmogorov length. Microscale phase change sustains small-scale inhomogeneity in turbulent clouds, evidenced by the trimodal distribution of cloud water gradients. For a given turbulence intensity, the microphysical relative dispersion scales cubically with the slope difference in the cloud water spectrum.

1 Introduction

Atmospheric clouds regulate the Earth's radiative budget and hydrological cycle, yet their internal structure remains one of the most difficult features to represent in weather and climate models. The key reason is the enormous range of scales involved. Cloud properties vary from the kilometer scale of convection to the sub-millimeter scale at which individual droplets condense and evaporate. An informative signature of this multiscale behavior is the appearance of a break in the energy spectrum of the cloud water field. This suggests that the power-law exponent in the Fourier space changes abruptly and that the regimes on either side of the break are governed by different physical processes.

A closer look at the mechanisms behind the break reveals that microscale phase change influences liquid water content (LWC) spectrum. Davis et al. (1999) experimentally measured the LWC in clouds at approximately 4 cm resolution. The corresponding energy spectra showed two distinct power-law regimes, separated by a break. They proposed that an additional variance source exists at small scales but it remained unidentified because of insufficient measurement resolution. In order to theoretically analyze the spectral break, Jeffery (2001) solved an advection-diffusion equation for the LWC with inclusion of phase change. The author found that a production subrange is created in the LWC spectrum by condensation and evaporation. The suggestion that microscale phase change injects variance at small scales is supported by Gotoh et al. (2021). They theoretically showed that the LWC spectrum exhibits two scaling ranges as cloud growth or decay modifies the spectrum. Recently, Wang et al. (2026) observed a physical-space break in the size distribution of growing clouds. Based on large-eddy simulations (LES), they concluded that the break originated from dynamical cloud growth.

A break in the scaling laws can be caused by the intersection of direct and inverse energy cascades. In a laboratory investigation, Xia et al. (2011) showed that energy is transferred from the small to large scales if a variance is externally imposed at the small scales. To distinguish between direct and inverse energy cascades, they calculated the third-order velocity structure function. Lilly (1983) suggested that Earth’s mesoscale atmosphere has more energy than predicted by theory and the source of extra energy is upscale energy transfer. This hypothesis was tested by Vallis et al. (1997) who performed large-eddy simulations of mesoscale convection and confirmed the presence of inverse energy cascade, although the upscale transfer was reportedly smaller than the downscale transfer. Cabanes et al. (2020) reported that planetary rotation inhibits vertical motion and creates a stratified atmosphere in which energy is transferred from the small-scale sources up to the large-scale jets.

While the abovementioned studies identified upscale energy transfer and consequent break in the scaling behavior, they did not explore how turbulence-cloud microphysics interactions affect these phenomena. This study examines the transfer of energy across scales within a turbulent cloud. We apply a particle-based direct numerical simulation (DNS) model which resolves all scales of the flow and tracks cloud droplets individually in Lagrangian fashion. Our model setup is such that it simulates entrainment-mixing between the cloudy and clear air. The grid resolution is much finer than that used in the previous studies of this problem. The objective is to analyze how microscale phase change acts as a localized source of variance and transfers energy from the small to large scales. By contrasting simulations with and without cloud droplets, we isolate the contribution of phase change. We also vary the intensity of flow turbulence and assess its impact on the energy cascades.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c\,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Above, $\boldsymbol{u}$ is the instantaneous air velocity vector, ν is the kinematic viscosity of air, p is the instantaneous air pressure, and ρ_a is the air density. $\boldsymbol{F}_c$ is an external force that maintains flow turbulence and is constructed in the Fourier space following the procedure of Eswaran and Pope (1988). The buoyancy force $\boldsymbol{F}_b$ is included but is too weak to sustain turbulence within a small domain. The conservation equations for the temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_h}{c_p}C_d + \mu_T\nabla^2 T + F_T\,, \tag{3}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -C_d + \mu_v\nabla^2 q_v + F_{q_v}\,. \tag{4}$$

Above, L_h is the specific latent heat, C_d is the time rate of exchange between water vapor and liquid water, and c_p is the specific heat of air. μ_T and μ_v are the molecular diffusivities while F_T, and F_{q_v} are the external forcing terms in the temperature and water vapor mixing ratio fields, respectively. The definitions of $\boldsymbol{F}_c$, $\boldsymbol{F}_b$, F_T, and F_{q_v} can be found in Sharfuddin et al. (2026). Note that F_T and F_{q_v} are necessary to sustain thermodynamic fluctuations in the respective fields. The environmental supersaturation (S_e) depends on the T and q_v, and is found as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97\frac{p_{sat}}{p - p_{sat}}} - 1\,, \tag{5}$$

where $q_{v,s}$ is the saturation vapor mixing ratio and p_{sat} is the saturation vapor pressure which can be calculated with respect to water as: (Huang, 2018)

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right). \tag{6}$$

The equations for the Lagrangian position ($\boldsymbol{X}$) and velocity ($\boldsymbol{V}$) vectors of cloud droplets are written as:

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t)\,, \quad \frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i}[\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)] + \boldsymbol{g}\,, \quad (\tau_p)_i = \frac{2\rho_l(r_i)^2}{9\rho_a\nu}\,. \tag{7}$$

where the subscript 'i' denotes the i-th particle, r is the particle radius, $\boldsymbol{u}_i$ is the instantaneous air velocity vector interpolated at the position of particle i, $\boldsymbol{g}$ is the gravity vector, τ_p is the particle response time, and ρ_l is the density of liquid water. Cloud droplets grow or shrink according to the following equation:

$$\frac{d\big(r(t)\big)_i}{dt} = \left(\frac{G}{r(t)} S_e(\boldsymbol{X}, t)\right)_i , \quad G = \frac{1}{\frac{L_h \rho_l}{k_T T}\left(\frac{L_h}{R_v T} - 1\right) + \frac{\rho_l R_v T}{\mu_v p_{sat}}} . \tag{8}$$

where G is the growth factor, k_T is the thermal conductivity in air, and R_v is the universal gas constant divided by the molecular weight of water. Then, C_d is determined as:

$$C_d(\boldsymbol{x}, t) = \frac{4\pi \rho_l G}{\rho_a a^3} \sum_{i=1}^{n} \big(r(t) S_e(\boldsymbol{X}, t)\big)_i , \tag{9}$$

where n is the number of cloud droplets within a grid cell having a volume of a^3. The curvature and solute effects are neglected in the present study because cloud droplets remain much larger than dry aerosols throughout the simulation period. Finally, the cloud water mixing ratio q_c is found as:

$$q_c(\boldsymbol{x}, t) = \frac{4\pi \rho_l}{3\rho_a a^3} \sum_{i=1}^{n} r_i^3(t) . \tag{10}$$

The q_c denotes cloud (liquid) water content and is directly affected by condensational growth or evaporative decay of cloud droplets.

3 Numerical Details

Flow turbulence is assumed to be homogeneous and isotropic. We generate three turbulent kinetic energy (TKE) dissipation rates by varying the band of forcing wavenumbers. We denote the three TKE levels as 'low' (L), 'medium' (M), and 'high' (H), respectively. The corresponding wavenumber bands are: $1 \leq k_f/k_{min} \leq 2.45$, $1 \leq k_f/k_{min} \leq 4.12$, and $1 \leq k_f/k_{min} \leq 13.86$ while the Kolmogorov length scales (η) are: 0.0019 m, 0.0015 m, and 0.00088 m. Note that $k_{min} = 2\pi/L = 12.27$ [1/m] is the minimum wavenumber and k_f is the forcing wavenumber. The external thermodynamic forcing terms (F_T and F_{q_v}) are constructed in the Fourier space with a wavenumber band of $1 \leq k_f/k_{min} \leq 4.12$. More details about external forcing and our particle-based DNS model can be found in Sharfuddin et al. (2026).

The physical model is a cube and has a volume of 0.512^3 m^3, with each side having $L = 0.512\ m$. The boundary conditions are triply periodic. The domain is equally divided into cloudy and clear-air regions at the start. A slab-like configuration is assumed so that the initial temperature and water vapor mixing ratio change sharply at the cloud/clear-air interface (Gao et al., 2018). The initial vapor mixing ratio is specified as:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & elsewhere \end{cases} . \tag{11}$$

Above, $q_v^{max} = 3.949\ g/kg$, $q_{v,e} = 0.3\ g/kg$, and $d \equiv L/2$ is the width of the cloud slab. q_v^{max} and $q_{v,e}$ represent cloudy (supersaturated) and dry (subsaturated) regions, respectively. The initial temperature field is prescribed as:

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle \big(q_v(z, t = 0) - \langle q_v(t = 0) \rangle\big) . \tag{12}$$

We simulate four cases: H, H0, M, and L. The details are provided in Table 1 along with the values of simulation parameters. 16×10^6 cloud droplets are initially placed in the cloudy region, giving a droplet number density of 119.2 cm^{-3}. Entrainment of dry air and turbulent mixing spread cloud droplets to all over the domain. The dominant phase change process is evaporation. The number of grid points is 256^3 and the grid resolution is 2 mm for all cases. We run simulations for a physical time of 20 s. The $k_{max}\eta$ value is 1.38 for Cases H and H0, 2.36 for Case M, and 2.98 for Case L, where $k_{max} = (256\pi)/L = 1570.27$ [1/m] is the maximum wavenumber. The $k_{max}\eta$ measures how accurately the viscous dissipation range is captured in a simulation and is prescribed to be at least greater than 1.0 (Pope, 2000).

Table 1

Details of the test cases

Case	TKE dissipation rate ϵ_{TKE} (m^2/s^3)	Level of momentum forcing $\boldsymbol{F_c}$	Initial radius of each cloud droplet (μm)	Stokes number $S_t = \tau_p/\sqrt{(\nu/\epsilon_{TKE})}$ at initial time	Inclusion of droplets
H	0.0056	High	15	$6.42e-2$	Yes
M	0.0006	Medium	15	$2.12e-2$	Yes
L	0.00025	Low	15	$1.36e-2$	Yes
H0	0.0056	High	15	N/A	No

Case	Description
H	high (H) level of flow turbulence, cloud droplets included
M	medium (M) level of flow turbulence, cloud droplets included
L	low (L) level of flow turbulence, cloud droplets included
H0	high (H) level of flow turbulence, cloud droplets excluded (0)

Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_h	2.5×10^6	J/kg	$\mu_T = \mu_v$	2.2×10^{-5}	m^2/s
ρ_l	1000	kg/m^3	c_p	1005.0	J/kg/K
$p(t=0)$	71446	N/m^2	$\langle T(t=0)\rangle$	270.75	K
ρ_a	0.92	kg/m^3	ν	1.5×10^{-5}	m^2/s
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K

4 Results and Discussion

This section presents the findings from our particle-based direct numerical simulations. The results for the TKE spectra as well as the variance (energy) spectra of temperature (E_T), water vapor mixing ratio (E_{q_v}), supersaturation (E_{S_e}), evaporation rate (E_{C_d}), cloud water mixing ratio

(E_{q_c}), and cloud droplet radius (E_r) are shown. The spectra are calculated using the fluctuating component of each variable. The scale-dependent skewness and spatial derivatives of the vertical velocity (w) and q_c are examined. The correlation between the spectral slopes in the E_{q_c} and the broadening of cloud droplet size distribution (CDSD) is also assessed.

The energy spectra are shown in Figures 1(a)-(f). The E_{TKE} is higher at the large scales for Cases H and H0. This is expected because only the low-wavenumber region is forced. The four E_{TKE} curves converge at the intermediate scales. Difference is visible in the dissipation range between high (H and H0) and medium/low (M/L) turbulence cases because $k_{max}\eta$ is smaller for the former. The TKE spectra (Figure 1(a)) are identical for Cases H and H0 (no cloud droplets), suggesting that cloud droplets do not affect the TKE. There appears to be a reduction of downscale transfer in the temperature and water vapor spectra at the small scales (Figures 1(b)-(c)) if cloud droplets are present (Case H, M, and L). Otherwise (Case H0), E_T and E_{q_v} decay smoothly. The disruption of direct cascade is more conspicuous in the E_{C_d} whose slope turns positive before the final decay (Figure 1(d)), and apparently an inverse cascade intersects with the direct cascade. This creates a break in the scaling behavior as the slope is negative (-2.91) to the left of the intersection (break) but is positive (1.35) to the right.

The causal driver of the inverse cascade is microscale phase change or evaporation in our model. Cloud droplets either grow or shrink in response to local supersaturation. This phase change event injects variance into the cloud water field at the microscale. McWilliams (1984) found that turbulence can organize small-scale fluctuations into larger coherent structures. In our simulations, this upscale organization manifests in the form of an inverse cascade as shown in Figures 1(d)-(f). Consequently, the spectrum of cloud water variance peaks in the high-wavenumber region or small scales (Figure 1(e)), where the inverse cascade is dominant. If k_b is the wavenumber corresponding to the break, the physical-space break occurs at wavelength $\lambda_b = (2\pi/k_b)$. Based on Figure 1(e), $\lambda_b = (2\pi/(10^{-0.2}/0.00088)\sim 8.76\ mm$, $\lambda_b = (2\pi/(10^{-0.15}/0.00153)\sim 13.48\ mm$, and $\lambda_b = (2\pi/(10^{-0.05}/0.0019)\sim 13.39\ mm$ for Cases H, M, and L, respectively. The value of λ_b/η is ~9.95 for Case H, ~8.8 for Case M, and ~7.05 for Case L. Hence, the break in the cloud water spectrum occurs between the inertial and dissipation regions (Jeffery, 2001) at $\lambda_b \sim 8\eta$. The break is directly related to the Kolmogorov scale, showing its dependency on the intensity of flow turbulence.

We see in Figure 1(e) that the right slope is positive (1.6) for all three cases while the left slope is found to be negative for Cases L (-0.72) and M (-0.5) but positive (0.68) for Case H. Stronger flow turbulence rapidly smooths out thermodynamic and microphysical gradients and reduces their variance (Sharfuddin et al., 2026). As a result, the E_T, E_{q_v}, E_{C_d}, and E_{q_c} contain less energy for Case H, as indicated by a smaller area under the curve in Figures 1(b)-(e). The reduction in the integral of E_{q_c} for Case H is mainly due to the weakening of the direct cascade at the large and intermediate scales. Hence, the direct cascade in the cloud water field is governed by flow turbulence while the inverse cascade is driven by microphysics. The radius spectra (E_r) in Figure 1(f) reaffirm the energy injection at high wavenumbers and upscale transfer. Also, a direct cascade is present in the low wavenumber region of E_r, as shown by local peaks at $k\eta \sim 10^{-1.7}$, $k\eta \sim 10^{-1.5}$, and $k\eta \sim 10^{-1.4}$ for Cases H, M, and L, respectively. Since the supersaturation and particle fields are coupled (Equation (8)), large-scale fluctuations in the S_e are transmitted to the r.

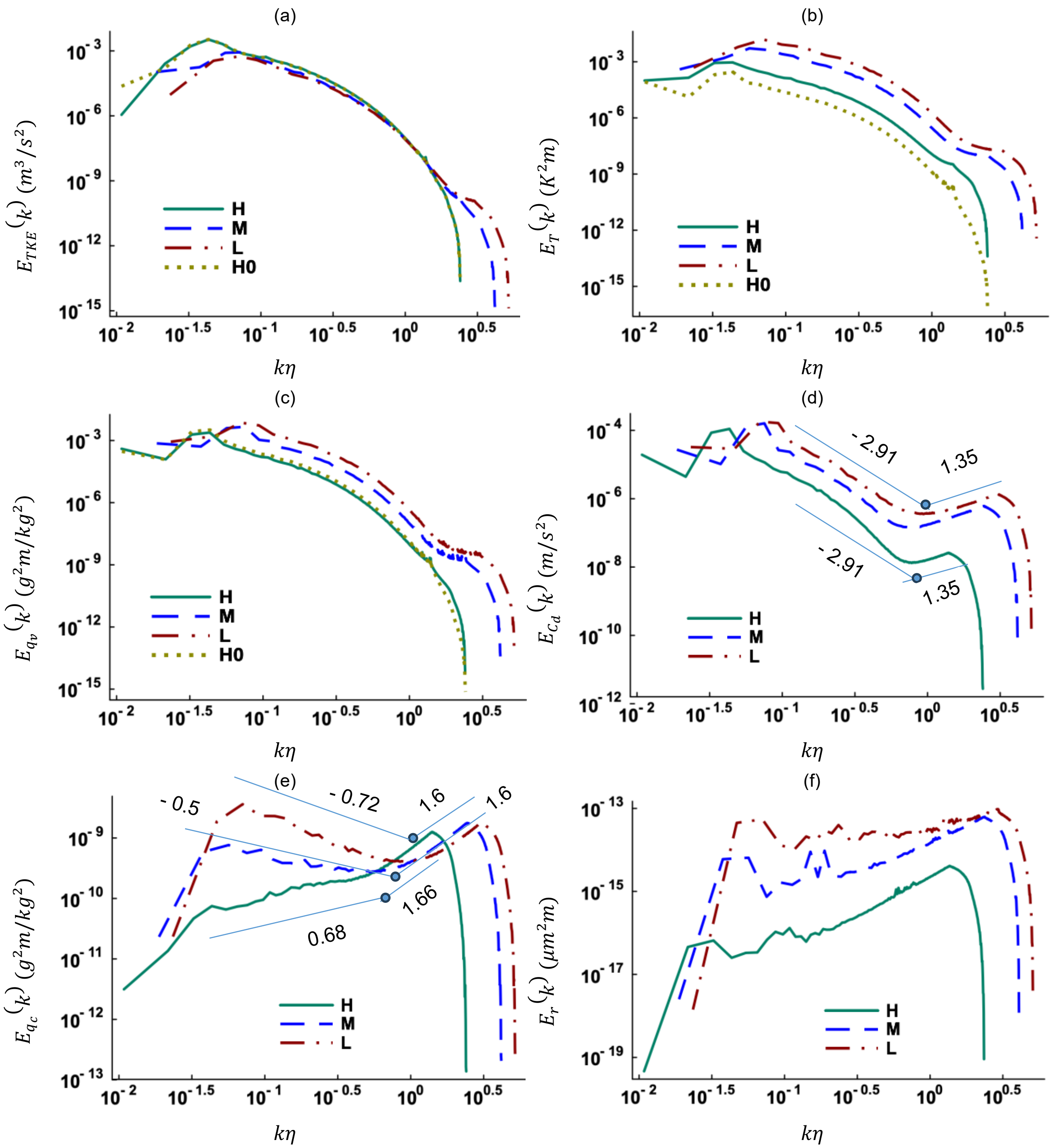

Figure 1. Energy spectra of the (a) turbulent kinetic energy (E_{TKE}), (b) temperature (E_T), (c) water vapor mixing ratio (E_{q_v}), (d) phase change term (E_{C_d}), (e) cloud water mixing ratio (E_{q_c}), and (f) particle radius (E_r) at 10 s. The intersection points in Figures 1(d)-(e) show spectral breaks.

The spectra in Figure 1 show how variance is distributed across scales, but do not reveal which way variance flows between scales. The scale-dependent skewness can be useful in this regard. Being the third order moment of increments, it vanishes for a Gaussian or symmetric field and becomes nonzero when nonlinear dynamics impose a directional asymmetry on the transfer.

The scale-dependent normalized skewness is defined, following the velocity-increment skewness factor of Davidson (2015) and generalized here to an arbitrary field ϕ, as

$$\gamma_{\phi}(\Delta z) = \frac{\langle\left(\phi(z+\Delta z) - \phi(z)\right)^{3}\rangle}{\langle\left(\phi(z+\Delta z) - \phi(z)\right)^{2}\rangle^{3/2}}. \tag{13}$$

where Δz is a finite vertical separation distance between two grid points. This normalized third-order structure function measures the asymmetry of increments and hence the direction of energy transfer. A zero value indicates no net transfer, while a negative sign implies a direct cascade and a positive sign suggests an inverse cascade (Frisch, 1995). We apply Equation (13) to test the signs of γ_w and γ_{q_c}, and thereby cascade direction. Note that the asymmetry imposed on these two fields arises from separate mechanisms. For the velocity field, vortex stretching amplifies fluid elements faster than compression weakens them and this asymmetry drives a downscale transfer (Davidson, 2015). For the cloud water field, the asymmetry is imposed by microscale phase change, which injects variance at the smallest scales and leads to upscale transfer.

We find that γ_w is highly negative at small separations and approaches zero as separation distance increases (Figure 2(a)). Therefore, energy flows steadily from the large to small scales, with no indication of upscale transfer in the momentum field. Ideally, γ_w should vanish at large separations as widely separated points become statistically independent. In our simulations, the domain length (L) is 0.512 m and the integral length scale l is approximated as 0.1 m, 0.15 m, and 0.2 m for Cases H, M, and L, respectively. Because L/l is smaller than the recommended value of 8 (Pope, 2000), velocity increments at large separations do not reach full statistical independence and a small residual skewness exists at large Δz. This residual does not affect the small-scale behavior, which is of interest in our study.

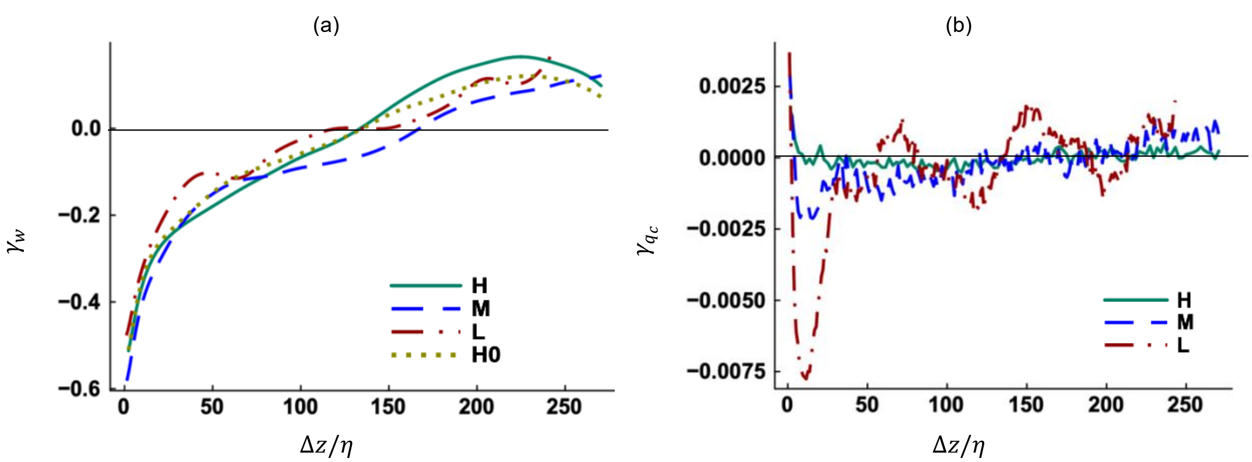

Figure 2. Scale-dependent skewness for the (a) z-velocity (γ_w) and (b) cloud water mixing ratio (γ_{q_c}) at 10 s.

The increments in cloud water content behave differently. Figure 2(b) shows that γ_{q_c} is slightly positive at a very small Δz for Cases H, M, and L. The positive skewness is consistent with the inverse cascade observed in the E_{q_c} (Figure 1(e)). As Δz increases, the direct cascade starts to dominate and γ_{q_c} becomes negative for Cases M and L before approaching zero at large separations. This is a physical-space confirmation of the transition between inverse and direct cascades. However, γ_{q_c} does not become meaningfully negative for Case H at any separation. This is again consistent with Figure 1(e) which shows that direct cascade is present for low and medium turbulence levels but not for the high turbulence level. The separation at which γ_{q_c} crosses the zero line (Figure 2(b)) is calculated as $\sim 1.62\eta = 3.1$ mm for Case L, $\sim 4.09\eta = 6.1$ mm for Case M, and $\sim 9.78\eta = 8.6$ mm for Case H. So, the crossover length is larger in response to high flow turbulence and reduced cloud water variance (Case H). We notice that the γ_{q_c} profiles are noisy while that for γ_w are smooth. The noise stems from the discrete, Lagrangian representation of cloud droplets.

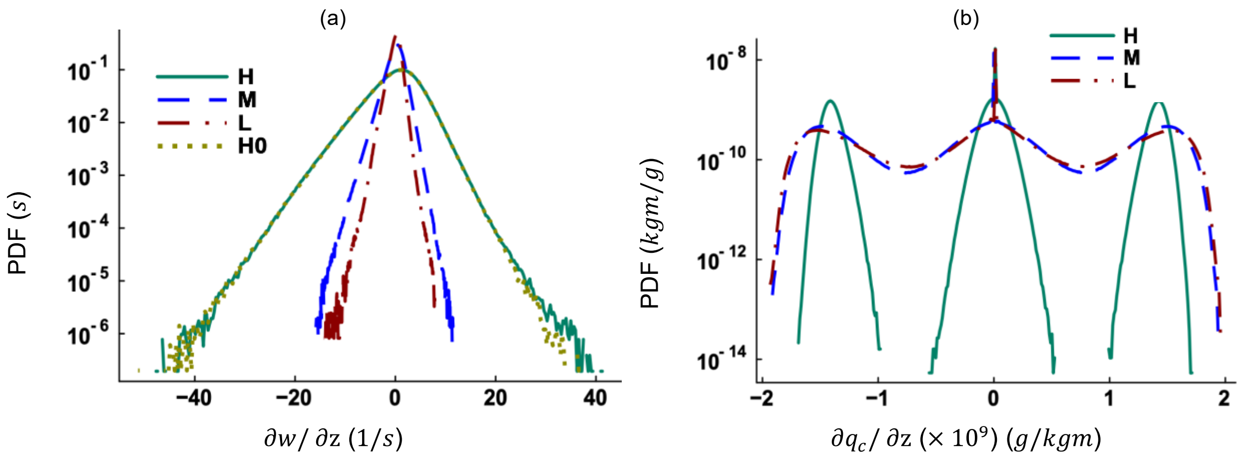

Figure 3. Probability density function (PDF) of the (a) vertical velocity derivative and (b) cloud water mixing ratio derivative at 10 s.

Spatial derivatives quantify the local steepness of a turbulent field. The derivatives are calculated in the z-direction using the central difference formula. Note that the PDFs are identical if the derivatives are calculated in the x- or y- coordinate direction. The vertical velocity derivative $\partial w/\partial z$ in Figure 3(a) suggests that strong turbulence (H and H0) extends the tail of the distribution because of more energy and intermittency. It also shows that the distribution is nearly symmetric, with equal magnitude of positive and negative gradients. The distribution of $\partial q_c/\partial z$ in Figure 3(b) is trimodal for all three cases (H, M, and L) permitting phase change. The simulation begins with equal volumes of cloudy and clear air, and subsequent mixing produces interleaved pockets of cloudy (supersaturated) and clear (subsaturated) air. The central peak corresponds to the interior of either pocket, where $\partial q_c/\partial z$ is essentially zero because q_c is nearly uniform in the cloudy pocket and zero in the clear air pocket. The left peak (negative $\partial q_c/\partial z$) marks the transition from a cloudy pocket outward into clear air, where the liquid water content falls sharply along the z direction. The right peak (positive $\partial q_c/\partial z$) marks the reverse transition from clear air into cloud. The $\partial q_c/\partial z$ varies gradually for Cases L and M, and in discrete jumps for Case H. Thus, the side peaks become more pronounced as turbulence intensifies. The sharp changes in q_c suggests the persistence of small-scale structures which inject variance at high wavenumbers. This reflects a basic difference between the velocity and cloud water fields. Molecular diffusion continuously smooths out velocity gradients, ensuring that the velocity derivatives stay finite and well-defined through a balance between turbulent production and diffusive decay. In contrast, the cloud water mixing ratio is constructed using discrete droplet positions (Equation (10)) and is not subject to molecular diffusion. The absence of diffusive smoothing allows the cloud water field to sustain variance at the small scales, fueling the inverse cascade responsible for the spectral break observed in Figure 1(e).

Table 2 reports the statistics of the slopes in E_{q_c} and the relative dispersion of cloud droplet radii $\sigma_r/\langle r\rangle$ at different times for high and low turbulence levels. It is evident that the higher the slope difference ($\Delta slope$), the larger the relative dispersion for Cases H and L. Between these two cases, the slope difference is higher for Case L at any observed time and so is the $\sigma_r/\langle r\rangle$. The right slope becomes increasingly positive for both cases (1.82 > 1.66 > 1.44 and 1.95 > 1.60 > 0.92) while the left slope becomes more positive for Case H but less negative for Case L. This systematic shift toward positive values suggests that turbulent homogenization weakens the direct cascade in cloud water variance spectrum and the relative contribution of small scales to cloud water variance increases. Again, $\sigma_r/\langle r\rangle$ increases with time for both turbulence levels and a positive correlation between small-scale fluctuations and relative dispersion appears to hold over time. We also observe that the relative dispersion scales roughly as the cube of the slope difference for a given turbulence intensity (Table 2). Let the ratio $(\sigma_r/\langle r\rangle)/(\Delta slope)^3$ be denoted by B. The values of B are 0.0577, 0.0574, and 0.0571 at 4 s, 10 s, and 18 s for Case H while the corresponding values for Case L are 0.0171, 0.0200, and 0.0173. Thus, B remains nearly constant within each case but varies depending on the turbulence level. The slope difference can therefore diagnose the broadening of the CDSD only if the turbulence intensity is explicitly known. Because no theoretical basis has been established for the cubic exponent, we report this scaling strictly as an empirical observation.

Table 2

Relationship between $\sigma_r/\langle r\rangle$ and slope difference in the E_{q_c}

Time (s)	Left slope	Right slope	$\Delta slope$	$\sigma_r/\langle r\rangle$	$B = \frac{\sigma_r/\langle r\rangle}{(\Delta slope)^3}$
Case H					
4	0.61	1.44	0.83	0.033	0.0577
10	0.68	1.66	0.98	0.054	0.0574
18	0.76	1.82	1.06	0.068	0.0571
Case L					
4	−0.88	0.92	1.80	0.10	0.0171
10	−0.72	1.60	2.32	0.25	0.0200
18	−0.61	1.95	2.56	0.29	0.0173

5 Implications and Concluding Remarks

The DNS results presented in Section 4 carry several implications for the modeling and observation of cloud microphysics, which are summarized below.

A) The upscale transfer of cloud water variance (Figures 1(e)-(f)) cannot be captured by the mesoscale and global models as their grid resolution is between 100 m to 100 km. Hence, these models need to parameterize the injection of small-scale variance in the cloud water field.

B) The scale-dependent skewness γ_{q_c} is positive at very small spatial separations and crosses the zero line at ~3 mm and ~9 mm for low and high turbulence levels, respectively (Figure 2(b)). Thus, resolving the intersection of direct and inverse cascades necessitates a grid spacing on the order of millimeters.

C) The trimodal distribution of $\partial q_c / \partial z$ (Figure 3(b)) suggests that cloudy and clear-air pockets exist within a cloud. Even though turbulent mixing is active, microscale phase change injects variance continuously and maintains sharp gradients. Such spatial inhomogeneity in clouds must be accurately represented in numerical models.

D) Table 2 shows that the relative dispersion of cloud droplet radii scales roughly as the cube of the slope difference in the cloud water variance spectrum for a given turbulence level. This relationship can be useful for sub-grid parameterization in large-scale atmospheric models.

We conclude that microscale phase change acts as a localized energy source within clouds and leads to the upscale transfer of cloud water variance. Capturing this phenomenon requires a particle-based DNS model which calculates phase change at precise droplet locations and prevents numerical diffusion from artificially destroying local gradients that fuel the inverse cascade. However, these conclusions rest on the idealized assumption of homogeneous and isotropic turbulence while turbulence in real clouds is inhomogeneous and anisotropic due to buoyancy and shear. Future iterations of this particle-based DNS model will implement non-periodic boundary conditions to account for buoyancy and shear effects and investigate energy cascades in realistic cloud turbulence.

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